

Database Cardinality

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| What is Cardinality?

Definition: Cardinality specifies the maximum number of instances of one entity that can be associated with an instance of another entity through a relationship.

Example: One customer can place how many orders? One order belongs to how many customers?

Cardinality helps determine:

- Foreign key placement
- Number of tables
- Whether a junction table is required
- Database constraints
- Business rules

| Types of Cardinality

Cardinality

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- ├— One-to-One (1:1)
- ├— One-to-Many (1:N)
- ├— Many-to-One (N:1)
- └— Many-to-Many (M:N)

| One-to-One (1:1)

Definition: One instance of Entity A is related to at most one instance of Entity B, and vice versa.

Example: Person and Passport.

Person	Passport
PersonID: 1, Name: aadya	PassportNo: P101, PersonID: 1

Implementation: Usually Foreign Key + UNIQUE constraint.

| One-to-Many (1:1)

Definition: One instance of Entity A can relate to many instances of Entity B. Each Entity B instance relates to only one Entity A.

Department	Employee
DeptID: 10, Name: CS	EmpID: 1, Name: arpit, DeptID: 10
	EmpID: 2, Name: aadya, DeptID: 10

Implementation: Foreign key goes on the many side.

| Many-to-One (N:1)

This is simply the reverse view of One-to-Many. Example: Many Employees → One Department. Same relationship, different viewpoint.

| Many-to-Many (M:N)

Definition: Many instances of Entity A can relate to many instances of Entity B.

Diagram: Student M ----- N Course

Implementation

Student	Course	Enrollment (Junction Table)
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StudentID: 101, Name: DBMS	CourseID: 101, Name: OS	StudentID: 101, CourseID: 101
		StudentID: 102, CourseID: 101

Enrollment contains: FK → Student, FK → Course. Primary Key: (StudentID, CourseID).

| Why are M:N relationships special?

In an M:N relationship, each entity can be associated with multiple instances of the other entity. A single foreign key cannot represent multiple values without violating normalization. The correct solution is to introduce a junction table such as Enrollment, which stores pairs of keys.